

Brac University

Department of Electrical & Electronic Engineering

Semester Spring-25

Course Number: EEE103L

Course Title: Computer Programming Laboratory

Section: 06

Lab Report



Experiment no.

Name of the experiment: Introduction to Algorithm and Flowcharts

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Lab report of Experiment 1:

Question 1:

Draw a flowchart for a program that calculates and prints the grade for a student based on their score. First ask the user for the obtained marks. Determine the grade based on the following grading scale:

90 or above: A

80 - 89: B

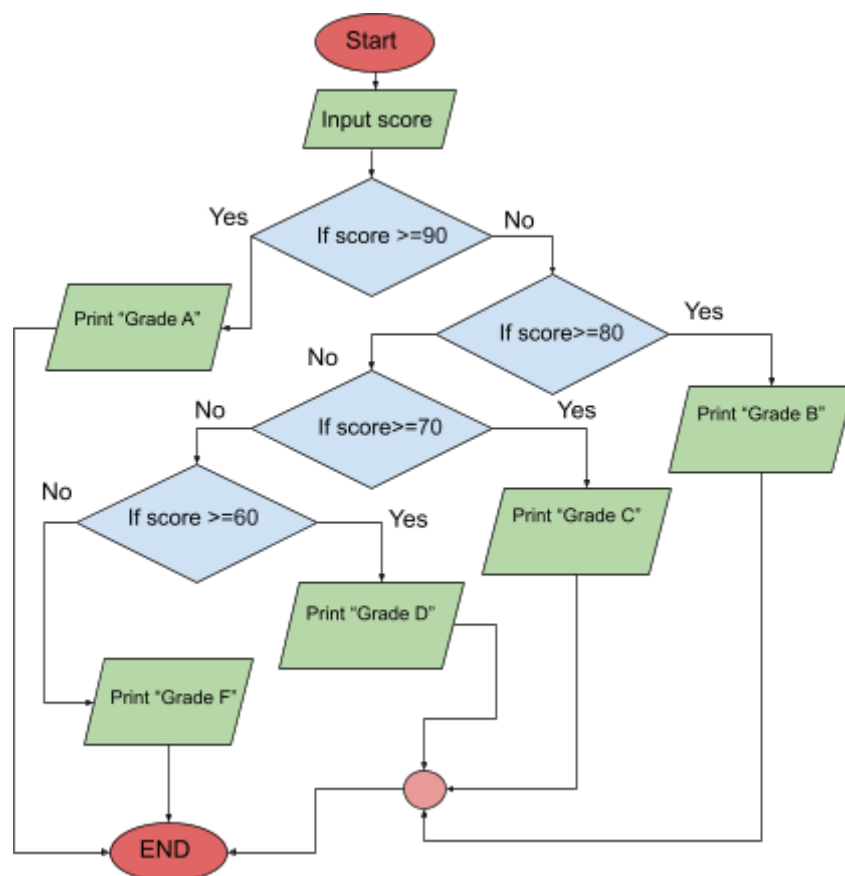
70 - 79: C

60 - 69: D

Below 60: F

Print the grade.

Flowchart:



Question 2:

Write an algorithm and draw the flowchart of a program that reads the radius of a circle and prints its circumference and area. If the area is larger than 100 print "Too large", if the area is smaller than 50 print "Too small"; otherwise print "Perfect".

Algorithm:

Step 01: Start

Step 02: Input the radius of circle **r**

Step 03: Calculate the circumference of the circle using the formula **$2 \times 3.1416 \times r$**

Step 04: Print circumference

Step 05: Calculate the **area** using the formula **$3.1416 \times r \times r$**

Step 06: Print **area**

Step 07: Check if **area** > 100 (units)

If yes; print "Too Large"

If no; then check if **area** < 50 (units)

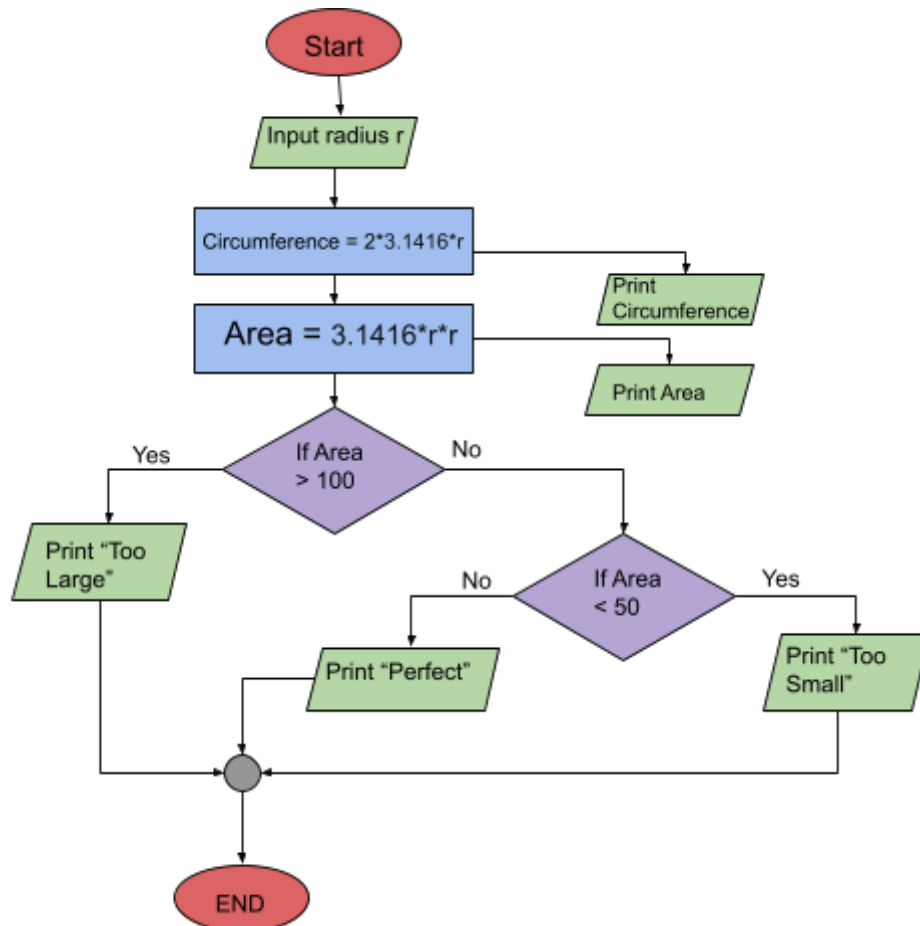
If yes; then print "**Too small**"

If no; then print "**Perfect**"

Step 08: Print the **area** of the circle

Step 09: End

Flowchart:



Question 3:

Write an algorithm and draw the flowchart of a program that reads five numbers as input from the user, prints whether the numbers are odd or even and prints the average of the numbers.

Algorithm:

Step 01: Start

Step 02: Set count= 0, sum =0

Step 03: If count < 5, go to step 4, else go to 10

Step 04: Input Number

Step 05: If num % 2 ==0, go to Step 6, else go to Step 7

Step 06: Print "Even"

Step 07: Print "Odd"

Step 08: Calculate : $\text{Sum} = \text{Sum} + \text{num}$

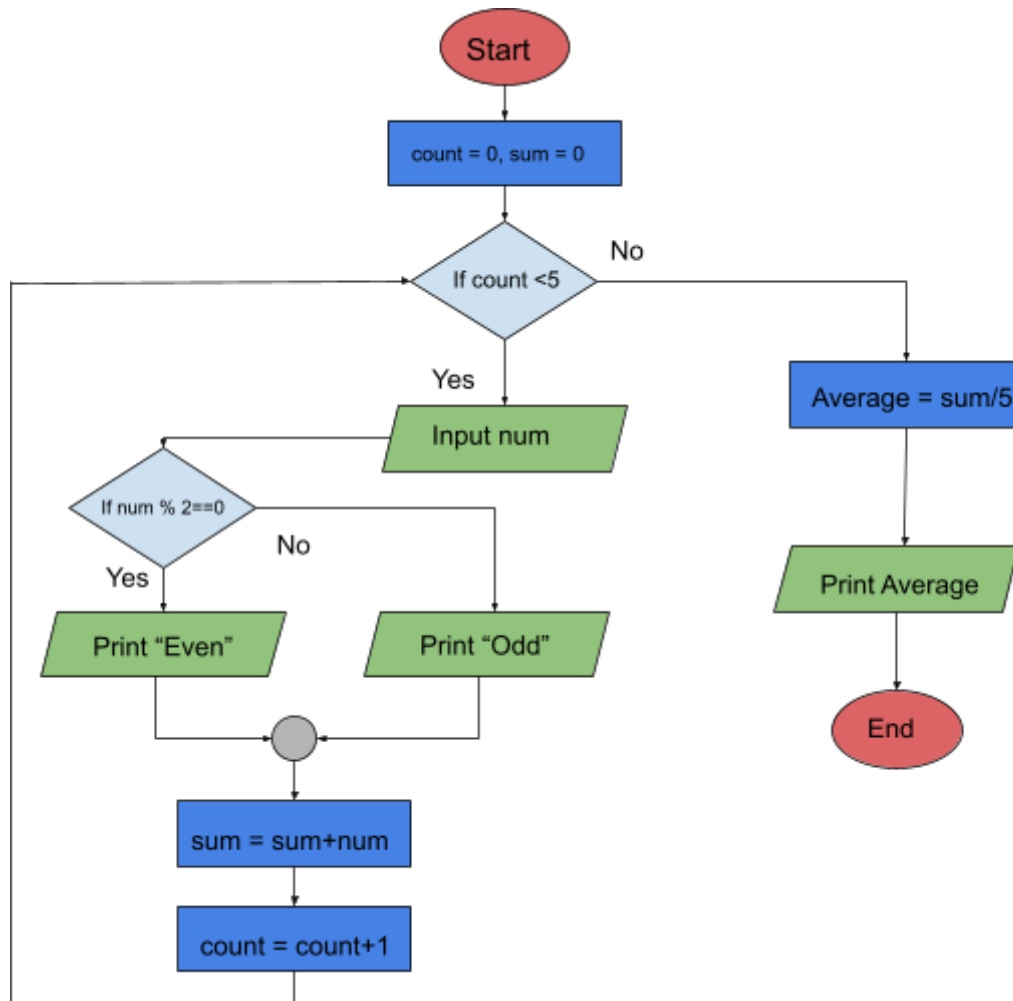
Step 09: Determine count by $\text{count} = \text{count} + 1$, go to step 3

Step 10: Calculate the average by $\text{average} = \text{sum} / 5$

Step 11: Print "Average"

Step 12: End

Flowchart:



Question 4:

Write an algorithm and flowchart of the following program: You are tasked with writing a program that calculates the total bill for a customer's purchase. The program should perform the following steps: Prompt the user to enter the prices of three items (item1, item2, and item3). Calculate the subtotal and add a 10% VAT to the total. If the total bill (including VAT) is more than 500, apply a 50 TK discount.

Print the total bill after applying the VAT and discount.

Algorithm:

Step 1: Start

Step 2: Input price of **item1**, **item2**, **item3**

Step 3: Sum the prices of the items using **item1+item2+item3**

Step 4: Find out 10% **VAT** on the sum using **sum*0.1**

Step 5 : Make the bill using **sum+VAT**

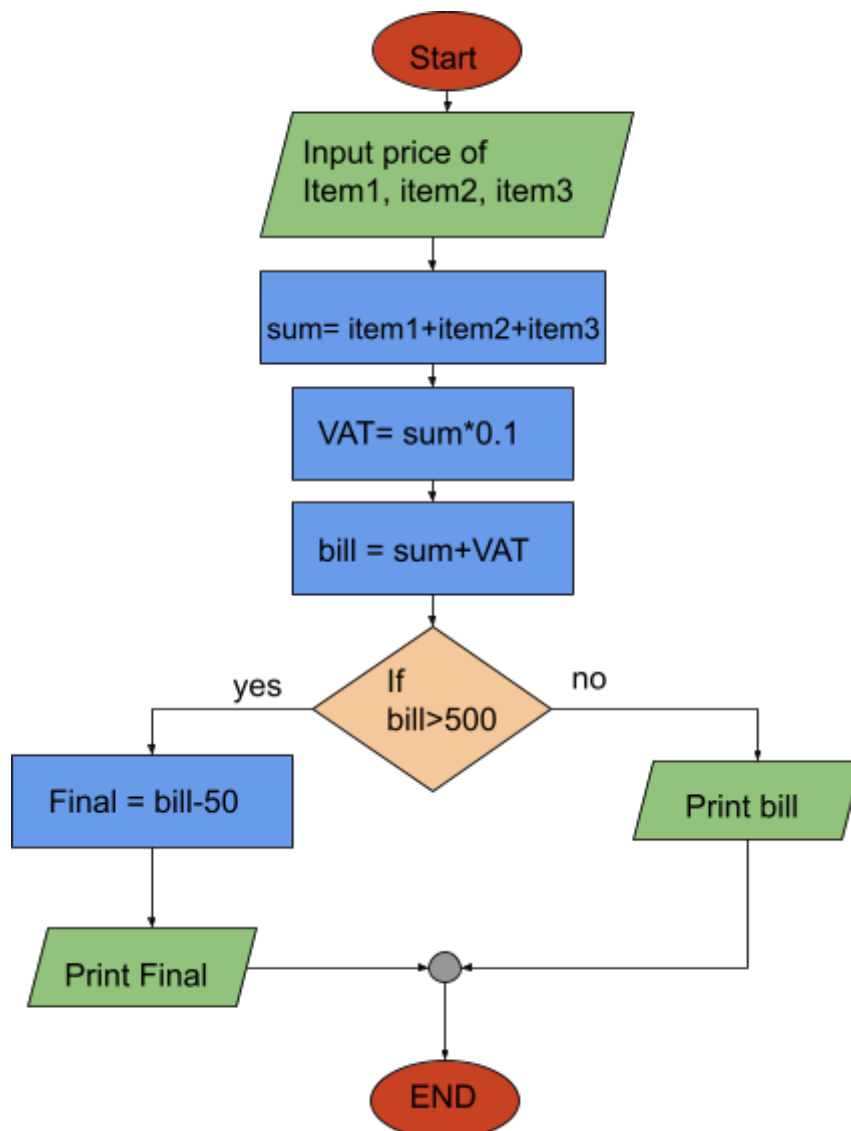
Step 6: Check if **bill>500**

If yes; calculate Final using **bill-50** and print **final**

If no; then print **bill**

Step 7: End

Flowchart:



Discussion of the experiment:

Algorithm: An algorithm is a set of step by step instructions that describes how to perform a task.

Flowchart: A flowchart is a visual representation of an algorithm to solve a problem or to perform a particular task.

We were able to understand the basic principles of identifying and sequence steps on how to solve basic problems. Step by step algorithms were written from start to end point on how a particular program will solve any given prompt or question. Necessary flowcharts were also drawn based on the algorithms to make the process of understanding easier. This will help us to plan, visualize and troubleshoot codes that we will run based on the created structure. Thus, the experiment of learning how to sequence algorithms and make flowcharts on them was a success.